Layers of Interpretation:
On Grammar and Compositionality

Emily M. Bender, Dan Flickinger, Stephan Oepen,
Woodley Packard & Ann Copestake

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Central claims

- Sentence meaning, but not speaker meaning, is compositional
- Systems attempting to understand speaker meaning would benefit from reusable, automatically derivable, task-independent representations of sentence meaning
- A compositional approach to creating sentence meaning representations provides
  - Comprehensiveness
  - Consistency
  - Scalability

Sentence meaning, speaker meaning

- Learning correlations between domain-typical surface forms and task-specific representations conflates:
    - timeless/conventional/standing/sentence meaning
    - utterer/occasion/speaker meaning
- Drawbacks:
    - resolving the same problems around grammatical structure for each task
    - unlikely to scale to general-purpose NLU

A meaning representation system is compositional if (working definition):

- it is grounded in a finite (possibly large) number of atomic symbol-meaning pairings
- it is possible to create larger symbol-meaning pairings by combining the atomic pairings through a finite set of rules;
- the meaning of any non-atomic symbol-meaning pairing is a function of its parts and the way they are combined;
- this function is possibly complex, containing special cases for special types of syntactic combination, but only draws on the immediate constituents and any semantic contribution of the rule combining them; and
- further processing will not need to destructively change a meaning representation created in this way to create another of the same type.

Benefits of compositionality: Comprehensiveness

- Grammar-based compositional approach ⇒ Every word and syntactic structure must be accounted for, or specifically deemed semantically void

- Narrower paraphrase sets, compare AMR (1), (2) (Banarescu et al 2014) to ERS (3)

(1) a. No one ate.
b. Every person failed to eat.

(2) a. The boy is responsible for the work.
b. The boy is responsible for doing the work.
c. The boy has the responsibility for the work.

Benefits of compositionality: Comprehensiveness

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- Narrower paraphrase sets, compare AMR (1), (2) (Banarescu et al 2014) to ERS (3)

(3) a. Kim thinks Sandy gave the book to Pat.
b. Kim thinks that Sandy gave the book to Pat.
c. Kim thinks Sandy gave Pat the book.
d. Kim thinks the book was given to Pat by Sandy.
e. The book, Kim thinks Sandy gave to Pat.

Benefits of Compositionality: Consistency

- Requiring meaning representations to be grounded in both the lexical items and syntactic structure of the strings being annotated significantly reduces the space of possible annotations

- Grammar based approach allows encoding of design decisions for machine application
    - Ex: arguments of *when*

- Human input still required, but choosing among representations is far simpler than authoring them
    - Development of grammar is still a big investment, but with big returns as the same grammar is applied over more and more text

Benefits of Compositionality: Scalability

- In amount of text annotated: Initial development of grammar pays off as it is applied to as much text as desired

- In genre diversity of the resource: One and the same grammar can be applied to texts from multiple different domains

    - Robustness techniques can compensate for lack of grammar coverage

- In the complexity of the annotations themselves: Grammar updates can be efficiently propagated across the treebank by reparsing corpus and rerunning annotation decisions (Oepen et al 2004)

    - Improve analyses of particular phenomena, or add layers of grammar-based annotation (e.g. partial constraints on information structure)

IAA Results

	Annotator Comparison			
Metric	**A vs. B**	**A vs. C**	**B vs. C**	**Average**
Exact Match	0.73	0.65	0.70	0.70
EDM_a	0.93	0.92	0.94	0.93
EDM_{na}	0.94	0.94	0.95	0.94

- Compare Banarescu et al (2013) triple-based IAA for AMR over web text of 0.71

The best of both worlds

- Grammar-based compositionally created meaning representations can only access information that can be implemented in a grammar

- But having comprehensive, integrated meaning representations is desirable (Basile et al 2012, Banarescu et al 2013), Ide & Suderman 2007)

- Proposal: Start from grammar-based meaning representations and process to produce larger paraphrase sets and/or add layers, such as fine-grained word sense annotations, additional coreference links, cross-sentential discourse relations